

How much of ANO's housing-estate vote goes with who lives there?

Census education and citizenship go with 43 % of ANO's 2025 estate gradient (H1b underpowered, by the design's collinearity rule); in matched precincts the gradient steepened between 2017 and 2025 (supported).

ANO, PRECINCTS WHERE MORE THAN SEVEN IN TEN FLATS ARE IN PANEL BUILDINGS	16 % where fewer than one in ten are · flats counted from the cadastre	26 %
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SHARE OF ANO'S ESTATE GRADIENT (PP OF VOTE PER 10 PP MORE PANEL FLATS, OTHER MEASURES HELD EQUAL) THAT GOES WITH EDUCATION AND CITIZENSHIP	estimate · range 31–44 % over 48 specifications · H1b underpowered	43 %
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ANO'S ESTATE GRADIENT, 2017 → 2025	pp per 10 pp of panel flats · change +0.26 (unrounded 0.545 → 0.801), range +0.24 to +0.30	0.55 → 0.80
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ANO AND DISTANCE TO A STATION, PER DOUBLING	90 % CI -0.40 to +0.05 · not supported: inconclusive	-0.17 pp
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OVERWHELMED? HERE'S THE SHORT VERSION

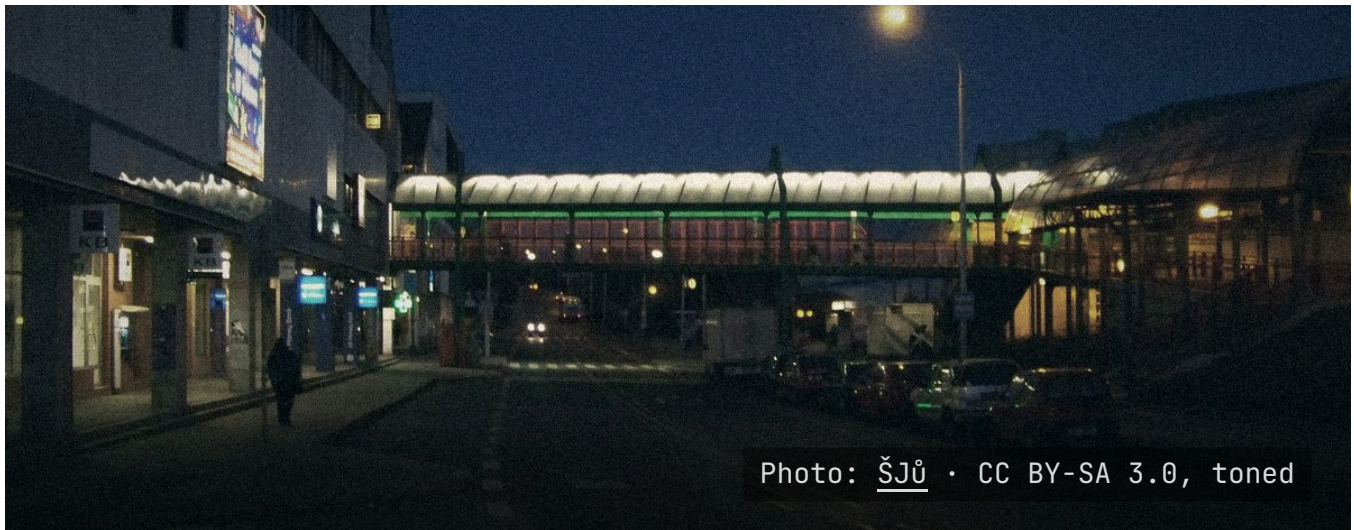
- ANO's vote is highest in the panel estates and rises with distance from the centre; the question is how much of the estate gap goes with who lives there.
- Census education and citizenship go with under half of that gap, and with about half once age, which may over-control, is added. The design's own rule for collinearity (a variance inflation factor above 10 in the education block) labels this test underpowered, so the share is an estimate, not a verdict on place against people.
- In older, matched precincts the gap grew between 2017 and 2025 (supported), which may partly reflect who moved in.

- The 43 % rests on 1 km census squares, 282 clusters in 56 districts; finer census data showing a much larger composition share would overturn it. The steepening uses no census data and rests on 819 matched precincts in 49 districts.
- The test on distance to a station is inconclusive (a negative estimate whose interval includes zero), the two turnout tests are uninformative, and nothing here describes how any resident votes.

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3

work since May 2024

+ About this study



Housing stock or residents

In the Chamber election of 3–4 October 2025 Prague voted SPOLU 33.9 %, ANO 19.9 %, the Pirates 16.9 % and STAN 13.4 %, on a turnout of 71.4 %¹, and across the city's 1 119 precincts ANO did best in the panel housing estates. Part 1 of this series followed the city council's votes; this part follows the voters. A panel estate is both a housing stock and a population: people who were allocated flats there in the 1970s and 1980s and grew old in them, their children, and later buyers of the cheapest flats in the city. ANO may do better in the estates because of the estates themselves (their distance, their shops, their neighbours) or because the estates hold more of the people who vote ANO anywhere. Electoral geography has argued about this for four decades, as context against composition²³, and for immigration attitudes in large European cities Maxwell, using the European Social Survey and the Swiss and German household panels, finds that city residents' more favourable views are mostly compositional, a result about attitudes, not votes⁴. Aggregate data cannot settle the question for individuals⁵⁶, but they can measure how much of the estate gradient (the points ANO gains for every 10 percentage points more flats in panel buildings, other measures held equal) is left once what the census knows about their residents is held equal. Prague adds a post-socialist case to that literature: its estates were built by the state and their flats allocated administratively rather than through a housing market, and the cadastre records the construction of

every residential building, so housing can be measured building by building. A follow-up study, its design written before modelling, therefore adds the 2021 census and the elections of 2017 and 2021 to the 2025 precincts and asks four things: how much of ANO's estate gradient goes with who lives in the estates, whether its association with distance to a station is small enough to be called negligible, whether the estates' vote has moved over eight years, and why turnout is lowest in the centre.

Data and the 2025 precinct map

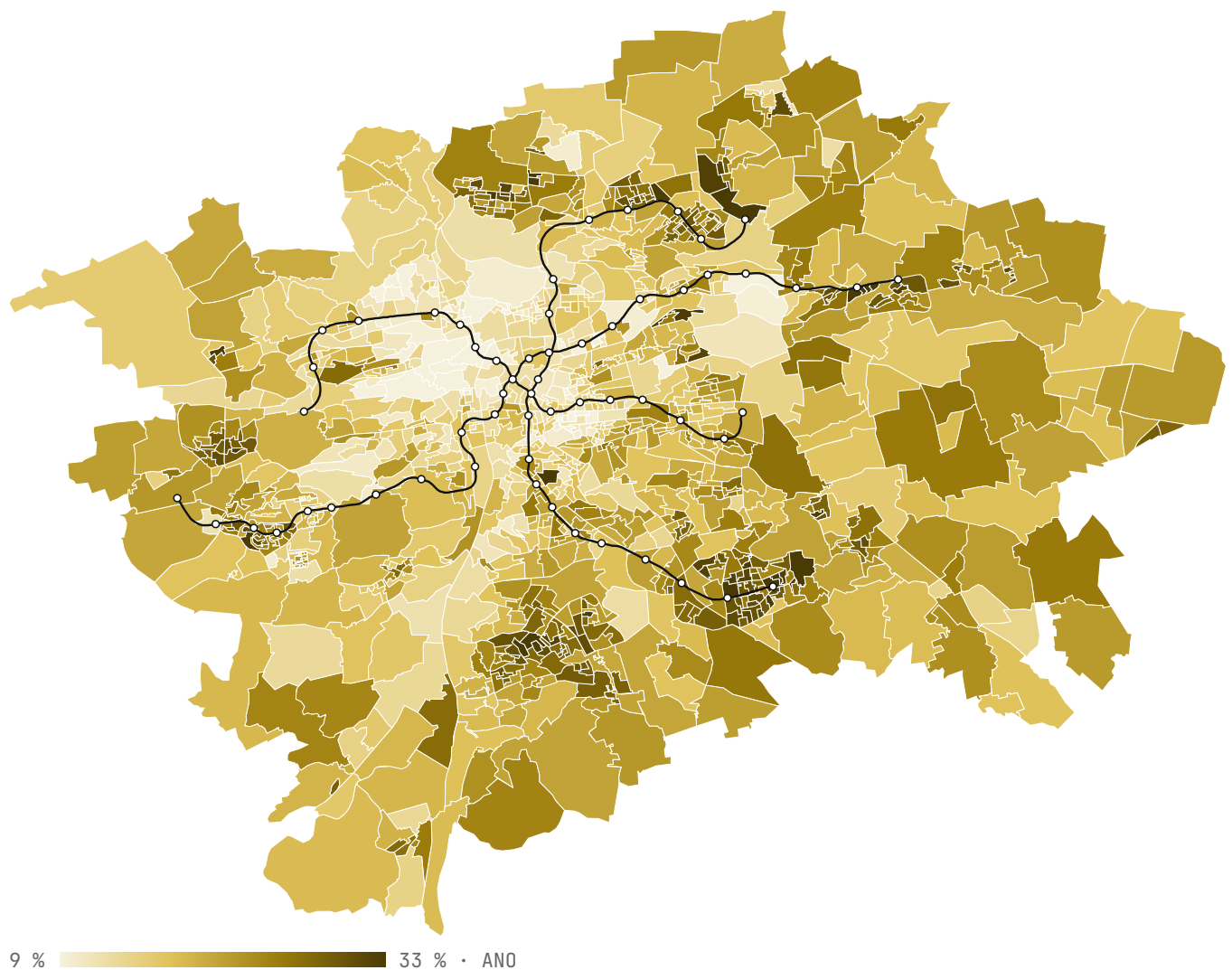


fig. 1 · Share of valid votes by precinct, Chamber election 2025, 1 119 precincts (ANO unless you pick another party, turnout or the share of panel flats above the map; hover or tap a precinct for its numbers); black: the metro and its 58 stations, Flora included (closed in 2026, open at the election); two of them, Pankrác and Českomoravská, were closed for reconstruction on election day and are kept, as in the registered distance. The darkest patches are housing estates: of the 40 precincts where ANO did best, 13 are in Praha 11 (Jižní Město) and most of the rest in Praha 8, 14, 12 and 18. Sources: ČSÚ, ČÚZK RÚIAN, PID GTFS, accessed 28 September 2026.

ANO's share rises with distance from the centre, from 13 % within two kilometres of Můstek to 25 % between eight and ten, then falls back to 22 % beyond ten, where the estates give way to family houses and villages inside the city boundary. The Pirates run the other way, from 22 % in the centre to 13 % at the edge, while SPOLU, the largest party, moves least, from 37 % to 31 % and back to 34 %. Three measurements can be put on every

precinct: how far its centre is from the nearest metro station, how far it is from Můstek, and what share of its flats are in panel buildings. Of the 735 755 flats in the precincts, 32 % are in panel buildings. Figure 2 lays the three side by side on one scale.

ANO's vote moves with the share of panel flats far more than with distance to the metro.

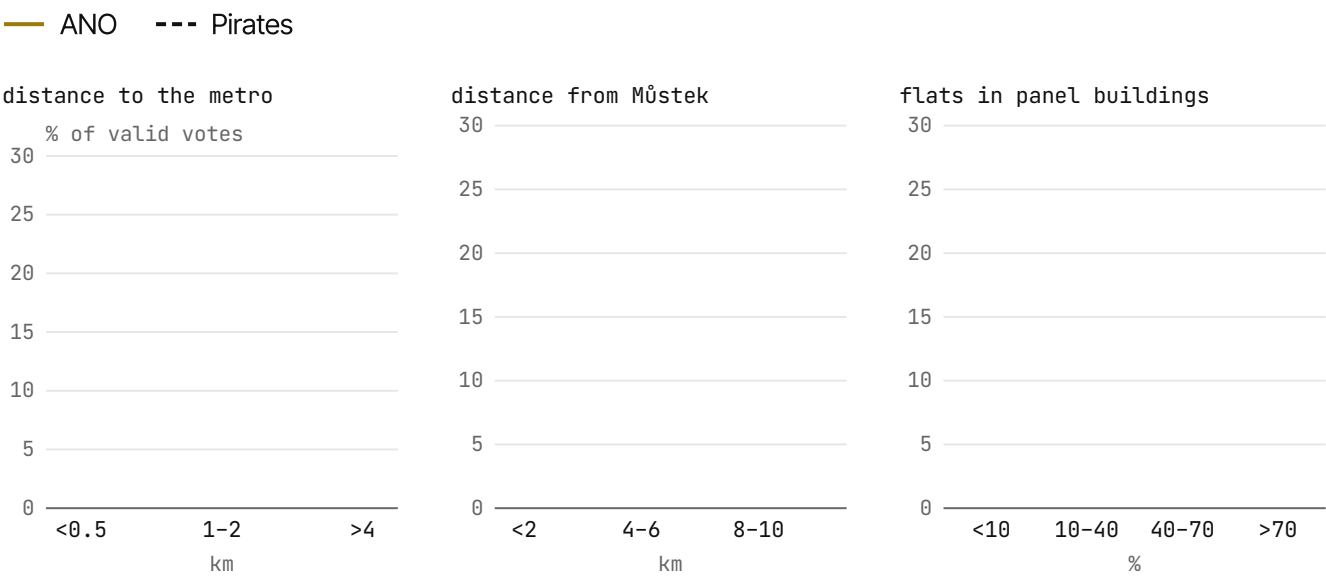


fig. 2 · ANO (ochre, solid line with dots) and the Pirates (dashed line with diamonds), % of valid votes, grouped by each measurement. Distances are from the precinct's centre; Můstek stands for the centre; a panel building is one the cadastre records as built of wall panels. Two precincts with no flats are left out (1 117 of 1 119 remain). Sources: ČSÚ 2025 precinct results; ČÚZK RÚIAN; PID GTFS; accessed 28 September 2026.

Near the metro or far from it, ANO's share stays between 19 and 23 %, and the small rise that exists comes with distance from the centre, probably because the stations thin out as the city does. Across the share of panel flats the change is much larger: where fewer than one flat in ten is in a panel building, ANO takes 16 % and SPOLU 37 %; where more than seven in ten are, ANO takes 26 % and SPOLU 29 %. A third of the city's flats are in panel buildings, so the gradient concerns a large part of the city rather than its margins.

In one regression weighted by valid votes, with the three measurements and the share of flats in family houses as a fourth, the metro's coefficient for ANO is -0.05 pp per doubling of the distance to the nearest station (95 % CI -0.28 to $+0.19$), which cannot be told from zero. Every ten percentage points of panel flats go with 0.9 pp more for ANO (0.8 to 1.0), and every doubling of the distance from Můstek with 2.0 more (1.5 to 2.4). The Pirates' share falls by 2.0 pp for every doubling of the distance from the centre and shows no measurable association with distance to a station either. The model's R^2 is 0.56 for ANO's share across precincts and 0.54 for the Pirates'. SPOLU loses 0.6 pp for every ten points of panel flats and gains 0.5 (0.4 to 0.7) for every ten points of flats in family houses, which may be why it recovers beyond ten kilometres, in the villa quarters and villages of its outer ring. The smaller parties follow the three measurements only weakly, with R^2 between 0.13 and 0.29. These 2025 estimates were known before the study below was designed.

The 2025 precinct counts add up to the Statistical Office's published result for Prague: 1 119 precincts, 898 829 registered voters, 632 938 valid votes, SPOLU 33.94 %, SPD 5.24 %¹. Boundaries come from the cadastre's current register, which keeps only the latest version of each precinct². Every record in it is dated after the election, most from two bulk updates in October and November 2025, and 22 were changed from December on;

dropping those 22 moves no estimate by more than a tenth of a point. The metro stations come from the public-transport timetable⁸, which lacks Flora, closed for reconstruction since 2 February 2026², so the station is added back as it ran on election day. A building's construction type is recorded for 98 % of the flats, and "panel" here means the cadastre's wall panels, concrete included.

The census counts people on a grid of 1 km squares¹⁰. Prague's median precinct is an eighth of a square, so every cell's counts were spread over the precincts it overlaps, in proportion to the flats standing in each on census day, from the cadastre's building register. This dasymetric allocation¹¹ puts 1 301 482 residents into Prague's precincts against the census's 1 301 432, and an independent allocation from the census's 958 basic settlement units in Prague agrees within 25 % for all but 13 precincts.

How this was done

The design fixes the questions, the hypotheses, the one test for each, the decision rules and fourteen robustness checks¹². It was written before modelling, but it first appears in the repository in the same commit as the results, so it is not independently timestamped. When it was written, the 2025 estimates above were known, and the author has worked with Prague's precinct and housing data continuously since May 2024, in work that led to this study. The plan sets the analysis in advance; it was written with knowledge of the data, not blind to them. Because the 2025 estimates were known, the metro hypothesis is framed as a test of whether the absence of a measurable association in 2025 holds up as a negligible one. Everything that changed afterwards is listed below.

A precinct gets only the composition of the square it sits in, while its housing is measured building by building. Comparing the two at different resolutions would flatter housing, so the main models add precincts up into the square that holds most of their flats, which gives 282 clusters, each measured at one resolution. The regressions weight by valid votes. Uncertainty comes from a wild cluster bootstrap by city district, which resamples whole districts so that neighbouring precincts are not treated as independent^{13,14}, with conventional, small-sample-corrected¹⁵ and spatial¹⁶ standard errors computed alongside. Each cluster belongs to the district that holds most of its flats, and the 282 clusters fall in 56 of Prague's 57 districts, so the effective sample for the composition and metro tests is 56 groups rather than 282 observations; the steepening test uses 819 precincts in 49 districts and the turnout tests 250 clusters in 44. The six confirmatory tests share a 5 % error rate through Holm's procedure¹⁷, which tests the smallest p-value against the strictest threshold and loosens it step by step.

The registered hypotheses and their results

Seven hypotheses were registered. H1a checks that the census measure of education behaves as expected and sits outside the confirmatory family of six. In the table, θ is the share of the estate gradient that goes with the composition block; for H1b the design registered three outcomes in advance: composition (θ significantly above one half), place beyond composition (θ significantly below one half and the remaining gradient significantly above zero) and indeterminate (neither). Its §5 adds a rule that overrides them: if any variance inflation factor of the realised design exceeds 10, H1b is reported as underpowered, not as place. At the design stage tertiary education reached 14.6 and vocational education 11.76, so H1b and H1b' carry the label underpowered, and θ is reported as an estimate (corrected on 9 October 2026; the first version applied the rule only to an indeterminate outcome).

Hypothesis	Test	Result
H1a · ANO does worse where more adults have a university or higher vocational degree check that the census measure works	coefficient < 0	supported -0.33 pp per pp, $p < 0.001$
H1b · education and citizenship go with more than half of ANO's estate gradient	$\theta > 0.5$	underpowered VIF 14.6 and $11.76 > 10$ (design §5) · $\theta = 0.43$, $p = 0.97$; reverse test $p = 0.03$
H1b' · the same for SPOLU's (negative) estate gradient	$\theta > 0.5$	underpowered same rule · $\theta = 0.42$, $p = 0.91$; reverse test $p = 0.09$
H2 · ANO's association with distance to a station is within ± 0.25 pp per doubling	equivalence (TOST: both ends of the 90 % interval inside ± 0.25)	not supported: inconclusive $p = 0.28$
H3 · ANO's estate gradient was steeper in 2025 than in 2017	interaction > 0	supported +0.26 pp per 10 pp, $p < 0.001$
H4a · the centre's turnout gap is smaller counted against residents than against the register	gradient smaller	uninformative $p = 0.80$, sign reversed
H4b · registrants who are not residents add less than one voter each	elasticity < 1 (% rise in envelopes per 1 % rise in the register)	uninformative 0.99 ± 0.02 , $p = 0.37$

p-values: one-sided, wild cluster bootstrap by district, 9 999 draws (56 districts for H1 and H2, 49 for H3, 44 for H4); H2 uses the largest of three standard errors. Holm thresholds run from 0.008 to 0.05; H3's p-value is below its threshold of 0.008. H1a is outside the family. H4a and H4b were registered as tests of the register and were relabelled uninformative after estimation, because the published register count includes voting-pass voters; their p-values are shown as computed.

How much of the estate gradient is composition

Across the 282 clusters, ANO's share rises by 0.99 pp for every ten percentage points more flats in panel buildings, holding the distance from the centre, the distance to the metro and the share of family houses equal. Adding what the census knows about education and citizenship lowers the gradient to 0.56 pp (95 % CI 0.47 to 0.66, bootstrap by district), and the census block raises R^2 from 0.70 to 0.87.

Education and citizenship go with 43 % of ANO's estate gradient; the registered test is labelled underpowered.

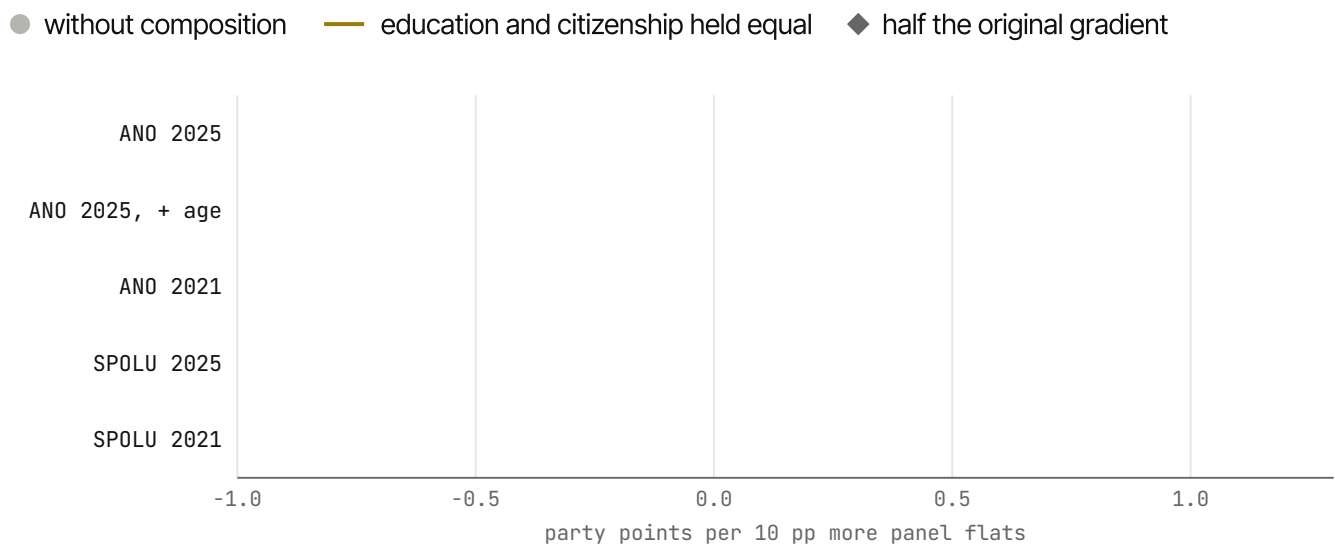


fig. 3 · The estate gradient, in party pp per 10 pp more panel flats, before (light grey) and after (arrow head: ochre for ANO, grey for SPOLU) holding education and citizenship equal; the mark on each row, a grey diamond (a dashed tick in the static figure), is half the original gradient, the line H1b had to cross. Right: the share that goes with composition. 282 grid clusters (280 in 2021), weighted by valid votes. Sources: ČSÚ elections 2021 and 2025, Census 2021 grid, accessed 28 September 2026.

A decomposition of the gap¹⁸ puts 0.44 of the 0.42 points on education and -0.02 on the share of foreign citizens, so the whole fall goes with education. Of the gradient, 57 % survives. The test in the other direction, whether more than half is left, gives $p = 0.03$ with the registered bootstrap and 0.04 with small-sample-corrected errors, but 0.06 with conventional and with spatial errors. Without the collinearity rule this would be the outcome "place beyond composition"; with it, H1b is labelled underpowered. Either way "place" would mean only "not going with education and citizenship": unmeasured income, tenure and length of residence could carry the remainder. For SPOLU the pattern is the same and weaker, with 42 % going with composition and neither direction significant, also labelled underpowered.

The census measures composition over a square kilometre, which is coarser than a street, and at registration this was expected to under-control for composition. The direction is not guaranteed here, because the census counts were spread over precincts in proportion to their flats, so the allocation error is correlated with the housing measure itself, and nothing here establishes its direction: 43 % may be understated or overstated. The registered block also leaves out age, because the estates' age structure is itself partly a product of the housing: flats were allocated to young families in one decade, and those families aged in place. With the share of over-65s and under-15s added, composition goes with 52 %, about half, and the test can no longer tell the answer from one half. The 2021 election, a replication specified in the design, gives 44 % for ANO and 31 % for SPOLU, and its registered outcomes disagree with 2025's: for ANO 2021 is indeterminate where 2025 was "place beyond composition", and for SPOLU 2021 is "place beyond composition" where 2025 was indeterminate. The design gives neither year priority.

Distance to the metro

The 2025 precinct model found no measurable association between distance to a station and the vote, and an interval around zero does not show that an association is absent, so the registered test asked the stronger question of whether the association, in the full model with composition, is small enough not to matter. The threshold was set at a quarter of a point per doubling of the distance to a station. Across the middle half of Prague's precincts that is half a point, a fortieth of ANO's vote in the city.

Every estimate of the association with distance to a station is small, but every interval reaches past the margin, so the test cannot call it negligible.

● estimate and 90 % interval ± 0.25: too small to matter

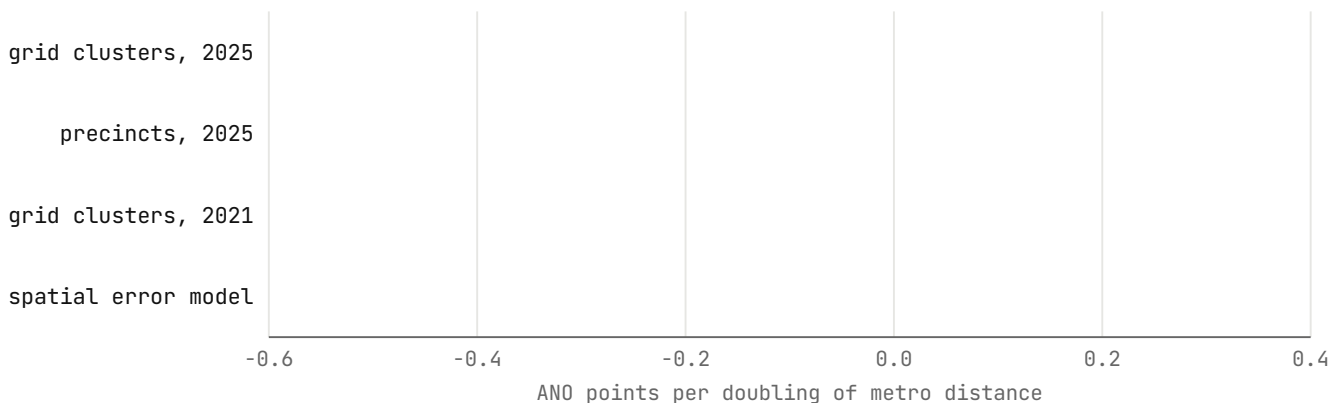


fig. 4 · ANO's coefficient on distance to a station in pp per doubling of the distance to a station, with its 90 % CI (the interval an equivalence test at 5 % uses), against the grey band declared too small to matter. 282 clusters (2025), 1 117 precincts, 280 clusters (2021). Sources: ČSÚ, Census 2021 grid, PID GTFS snapshot of 28 September 2026.

Every estimate is small and every interval crosses zero, but every interval also reaches past -0.25 . The design-stage power check, using a pessimistic residual variance, expected a standard error of about 0.31, above the margin, and so a test with almost no power. The realised standard error is 0.14, so the test could have passed with an estimate near zero. It does not, because the estimate itself, -0.17 , lies close to the -0.25 margin and its 90 % interval reaches -0.40 . The margin was not widened afterwards, because the 2025 estimate was already known and a margin chosen after seeing it would prove nothing. The test is therefore inconclusive: the estimate is negative and its interval includes zero, so no direction is established. Distance to a station is not assigned at random either. The lines were built to serve estates and centres, so the coefficient also carries the age of the housing, the shops and services around stations, any effect of distance from the centre beyond the logarithmic form used, and who chose to live near a line; the model holds only the centre distance, the housing shares and the census block equal.

ANO's estate gradient in 2017, 2021 and 2025

Precinct boundaries move between elections and the cadastre keeps only the current ones. The comparison over time therefore uses the 819 precincts that exist under the same number in 2017, 2021 and 2025 and whose register grew or shrank no more than 10 % beyond what new flats inside them predict. Matched precincts have

more panel flats than the rest (40 % against 27 %), lie slightly further from the centre on average, and gave ANO more of their vote in 2025 (20.5 % against 18.2 %), probably because the unmatched are mostly where the city is still building, so the comparison describes the older city rather than all of it.

ANO's estate gradient steepened from 0.55 in 2017 to 0.80 in 2025.

— 819 matched precincts ■ 5 % (410) and 20 % (982) matching rules

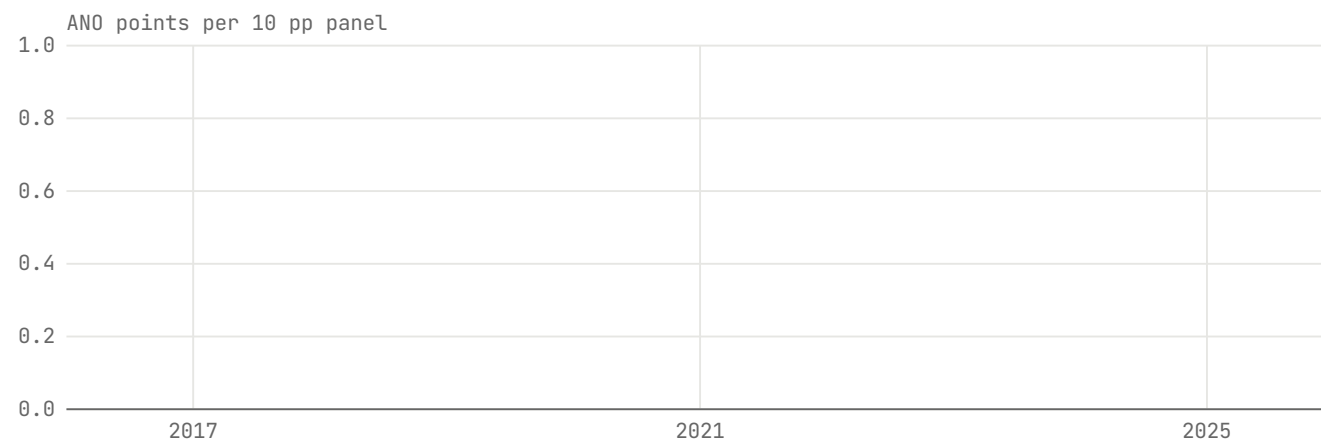


fig. 5 · ANO's estate gradient in pp per 10 pp of panel flats, by election, 819 matched precincts (ochre line); light ochre band: the same with the 5 % (410 precincts) and 20 % (982) matching rules. Distance from the centre, distance to the metro and family-house share are held equal in each year. Sources: ČSÚ elections 2017, 2021, 2025; ČÚZK RÚIAN; accessed 28 September 2026.

Between 2017 and 2025 ANO's estate gradient grew by 0.26 pp per 10 pp of panel flats ($p < 0.001$), from 0.55 to 0.80 (0.545 and 0.801 unrounded), with 2021 in between. It passes its Holm threshold, and it holds under the stricter and the looser matching rule (+0.24 and +0.30) and with weights that correct for which precincts could be matched (+0.26). The model holds housing fixed, applying today's housing stock to all three elections, but not people, since the census was taken once, between the elections, so the result shows that ANO's vote became more concentrated in the estates without showing whether the estates or their residents changed.

Turnout in the centre

The published register count per precinct includes voters who came with a voting pass, and passes are used most where people are away from home: in all 14 819 precincts in the file, abroad included, envelopes never exceed that count, which is what one would see if pass voters are added to it. This was found only after the data were joined, and because the published files do not allow the register and the pass voters to be separated, both turnout tests are uninformative. In 2025 turnout was lowest in the centre, 66.8 % within two kilometres of Můstek against about 73 % from four kilometres out, and it barely changed with the share of panel flats (71–72 % across the four groups). The registered explanation was that the centre has more voters who are registered there but live elsewhere, since Czech permanent residence is an address in a register and people with no address of their own are registered at the district office. With the 158 precincts that contain a district office¹⁹, a hospital²⁰, a care home²¹ or a prison²² left out, turnout counted against the register rises by 1.9 pp per doubling of the distance from Můstek, while counted against the census's resident Czech adults it falls by 4.3, and envelopes scale with the register at an elasticity (the % rise in envelopes for a 1 % rise in the register) of 0.99, with residents adding nothing once the register is known. With the flagged precincts kept, as the design registered for reporting, the

elasticity is 0.74 ($p = 0.01$), equally uninterpretable for the same reason. Because pass voters are inside the register count, neither result can be read as evidence for or against the registration explanation.

Robustness checks

In all 48 specifications composition goes with 31–44 % of the gradient; in 36 of them significantly less than half, in 12 neither.

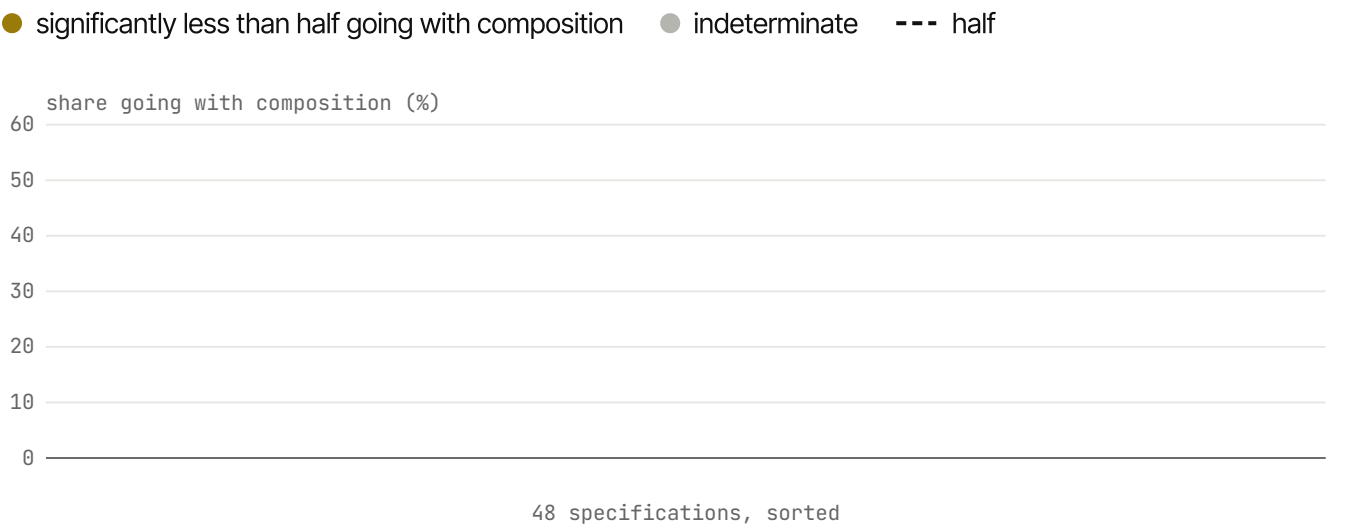


fig. 6 · Share of ANO's estate gradient that goes with education and citizenship (the registered block, without age) in 48 specifications: weighted or not, allocation by flats or by area, panel defined narrowly or broadly, all clusters or large precincts only or homogeneous squares only, bootstrap or spatial errors. Ochre: significantly less than half (36 of 48). Light grey: indeterminate (12). None: more than half. Sources as fig. 3.

Check	ANO θ	SPOLU θ	metro
main	43 %	42 %	−0.17
unweighted	34 %	36 %	−0.20
without precincts redrawn after Dec 2025	43 %	44 %	−0.16
precincts with 300+ voters	43 %	42 %	−0.17
panel: wall panels only	43 %	44 %	−0.18
census spread by area	40 %	36 %	−0.14
education stated, 50+ residents	42 %	40 %	−0.18
composition of citizens only	42 %	42 %	−0.19
allocation check passed	41 %	39 %	−0.15
homogeneous squares only	44 %	38 %	−0.43

Further check	Result
48 specifications of the registered block (fig. 6)	ANO θ 31–44 %
leave one district out	θ moves by at most 0.02
fractional logit, for shares between 0 and 1 ²³	ANO θ 46 %
single precincts (composition coarser than housing)	ANO θ 30 %
spatial error model; residual Moran's $I = 0.06$	same answers, Hausman test ²⁴ $p = 0.62$
panel shares shuffled within districts	attenuation 0.19 pp, observed 0.42
metro test with the 56 stations open on election day, without Pankrác and Českomoravská (post hoc, not registered)	−0.16 pp per doubling, 90 % CI −0.39 to 0.06, inconclusive

Across the 48 specifications composition goes with between 31 and 44 % of ANO's estate gradient under the registered block. Single precincts give a lower share, presumably because their composition is coarser than their housing, which inflates the leftover gradient by construction. Shuffling panel shares within districts leaves an apparent attenuation of 0.19 pp rather than zero, presumably because districts differ in both housing and people, and the observed 0.42 lies outside that range.

What this means

The results locate ANO's vote in the housing estates and measure how much of it goes with who lives there, without identifying any effect of the estates on anyone's vote. Measured are the votes, the housing and the distances: ANO takes 26 % where more than seven flats in ten are in panel buildings and 16 % where fewer than one in ten are. Estimated, from 1 km census squares, is the share of that gradient going with education and citizenship: 43 % (31–44 % across 48 specifications), with the direction of its error unknown, about half once age is added, and a test the design's collinearity rule labels underpowered. The rest cannot be credited to the estates as places, since income, tenure and length of residence are unmeasured. In the older, matched precincts the gradient rose by 0.26 pp per 10 pp of panel flats between 2017 and 2025 (+0.24 to +0.30 across matching rules), with residents not held fixed. The association with distance to a station, -0.17 pp per doubling (90 % CI -0.40 to $+0.05$), has no established direction, and the turnout tests are uninformative.

What this does not show

No result here says how any resident votes, because every measurement is of areas⁵. The census is from March 2021, four and a half years before the 2025 election, taken during distance teaching and before the refugees of 2022. The grid is coarse, which limits what composition can capture. The areas are Prague's, and a different grid would give somewhat different numbers²⁵. The leftover estate gradient is what the census does not measure, and the place itself is only one candidate for it: income, tenure, how long people have lived there, the age of the buildings and the shops and services nearby are not in the model, and any of them could carry it. The association with distance to a station is open to the same confounders, since stations were built where estates and centres were planned. The steepening since 2017 describes the matched, older precincts, with residents' composition not held fixed over time. None of the estimates is a causal effect of living in a panel building.

Reproduction

The analysis runs from `tools/praha/` in the site's repository. Each script's header gives its `uv run` command with its dependencies (pandas, numpy, scipy, statsmodels, shapely, pyproj, pyarrow, matplotlib); no lock file pins their versions yet. Raw downloads are cached in `tools/data/praha2/` and `tools/data/praha2x/`, which are not committed, and the hashes of the files present at registration are in [prague-rings-files.sha256](#). The scripts run in this order:

- [metro.py](#): the 2025 description and the map data
- [rings_data.py](#): the design frame, before any result is joined
- [rings_power.py](#): the design-stage checks

- [rings_extended.py](#): the registered tests, written to `assets/praha/rings_extended.json`
- [rings_robust.py](#): the robustness checks, written to `assets/praha/rings_robust.json`
- [rings_metro_open.py](#): the post-hoc metro test with the stations open on election day, written to `assets/praha/rings_metro_open.json`
- [figures_metro.py](#) and [figures_rings_extended.py](#): the figures

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